

Definition 12.1

A *transposition* in S_n is a cycle (i_1, i_2) of length 2.

Theorem 12.2

Every permutation in S_n can be written as a product of transpositions.

Theorem 12.3

Let $\alpha \in S_n$ and let $\alpha = \beta_1 \circ \beta_2 \circ \cdots \circ \beta_r$ be a decomposition of α into a product of r transpositions.

- If r is even, then every other decomposition of α into transpositions consists of an even number of transpositions.
- If r is odd, then every other decomposition of α into transpositions consists of an odd number of transpositions.

Lemma 12.4

Let $\beta_1, \dots, \beta_r$ be transpositions in S_n such that

$$\beta_1 \circ \beta_2 \circ \cdots \circ \beta_r = \varepsilon$$

where ε is the identity permutation. Then r is an even number.

Proof of Theorem 12.3.

Definition 12.5

A permutation $\alpha \in S_n$ is *even* if it can be written as a product of even number of transpositions and it is *odd* if it can be written as a product of an odd number of transpositions.

Theorem 12.6

The subset of S_n consisting of all even permutations is a subgroup of S_n .

Definition 12.7

The subgroup of S_n consisting of even permutations is called an *alternating group on n letters* and it is denoted by A_n .

Theorem 12.8

For $n \geq 2$ the alternating group A_n has order $\frac{n!}{2}$.

Definition 12.9

The *sign* of a permutation $\alpha \in S_n$ is defined as follows:

$$\text{sign}(\alpha) = \begin{cases} +1 & \text{if } \alpha \text{ is even} \\ -1 & \text{if } \alpha \text{ is odd} \end{cases}$$